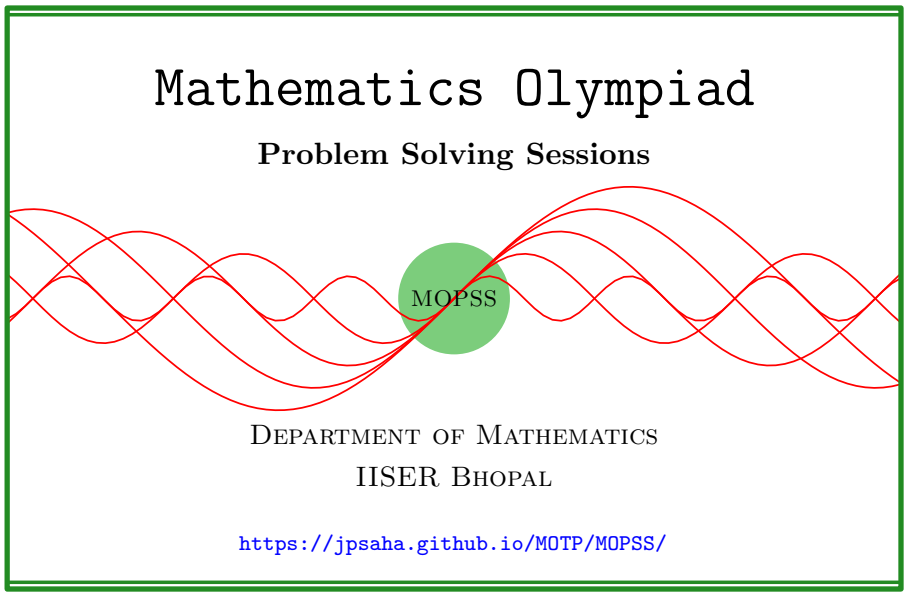
Pigeonhole principle

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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§1 Pigeonhole principle

See [Sob13, Chapter 2], [Mat24, Chapter 6].

Example 1. Show that in any group of six people, there are either three mutual acquaintances or three mutual strangers.

Remark. This is a special case of Ramsey's theorem, which states that for any two positive integers m and n, there exists a least positive integer $R(m,n)$ such that any graph on $R(m,n)$ vertices contains either a clique of size m or an independent set of size n. The number $R(m,n)$ is called the Ramsey number. The problem above states that $R(3,3)=6$.

Remark. Note that

$$2^2 = 4,$$
$$2^6 = 64,$$
$$2^5 = 32,$$
$$2^{25} = 33554432$$

holds.

Example 2. Are there two powers of 2 such that the blocks of their last three digits are the same?

Example 3. Let $X = \{a_1, a_2, \ldots, a_5\}$ where $a_1, a_2, \ldots, a_5$ are integers which are perfect squares. Show that there exists a subset $Y = \{b_1, b_2, b_3\}$ of X such that $b_1 + b_2 + b_3$ is divisible by 3.

Solution 1. Since perfect squares are congruent to 0 or 1 modulo 3, at least three elements b_1, b_2, b_3 of X are congruent to either 0 modulo 3, or 1 modulo 3. So their sum is divisible by 3. ■

Exercise 4 (RMO 1990 P1,). Two boxes contain between them 65 balls of several different sizes. Each ball is white, black, red or yellow. If you take any 5 balls of the same colour at least two of them will always be of the same size (radius). Prove that there are at least 3 balls which lie in the same box have the same colour and have the same size (radius).

> **Walkthrough** —
>
> **(a)** At least how many balls are there in the box with more balls?
>
> **(b)** At least how many balls are there in a **large** colour class in this box?
>
> **(c)** What about the sizes of the balls of this colour class?

Solution 2. By the pigeonhole principle, one box contains at least 33 balls. Applying the pigeonhole principle once again, it follows that at least 9 balls in this box are of the same colour. Since among five balls of the same colour, at least two of them are of the same size we conclude that these 9 balls of the same colour are of at most four different sizes. By the pigeonhole principle, at least three of these 9 balls are of the same size. ■

Exercise 5 (RMO 1996 P7,). If A is a fifty-element subset of the set $\{1, 2, 3, \ldots, 100\}$ such that no two numbers from A add up to 100, show that A contains a square.

> **Walkthrough** — Decompose $\{1, 2, \ldots, 100\}$ as the union of the subsets $\{1, 99\}$, $\{2, 98\}$, $\ldots$, $\{49, 51\}$, $\{50\}$, $\{100\}$. Does one of these subsets contain two perfect squares?

Solution 3. Note that the set A is formed by choosing at most one element from each of the following sets $\{1, 99\}, \{2, 98\}, \ldots, \{49, 51\}, \{50\}, \{100\}$. If 100 lies in A, then we are done. Otherwise, A is formed by choosing at most one element from the first 50 sets. Since A has 50 elements, one element of each of these 50 sets lies in A. In particular, one of $36, 64$ lies in A, and hence A contains a perfect square. ■

Exercise 6 (Putnam 2002 A2). Given any five points on a sphere, show that some four of them must lie on a closed hemisphere.

Remark. Let $\mathcal{S}$ be a sphere and $\mathcal{C}$ be a great circle on it. Then $\mathcal{C}$ divides $\mathcal{S}$ into two parts, which are called the *hemispheres defined by* $\mathcal{C}$. Any such hemisphere together with the great circle $\mathcal{C}$ is called a *closed hemisphere*. In **??**, there are a few examples of closed hemispheres. Those are the grey ones together with the great circles marked in red, and the blue ones together with the great circles marked in red.

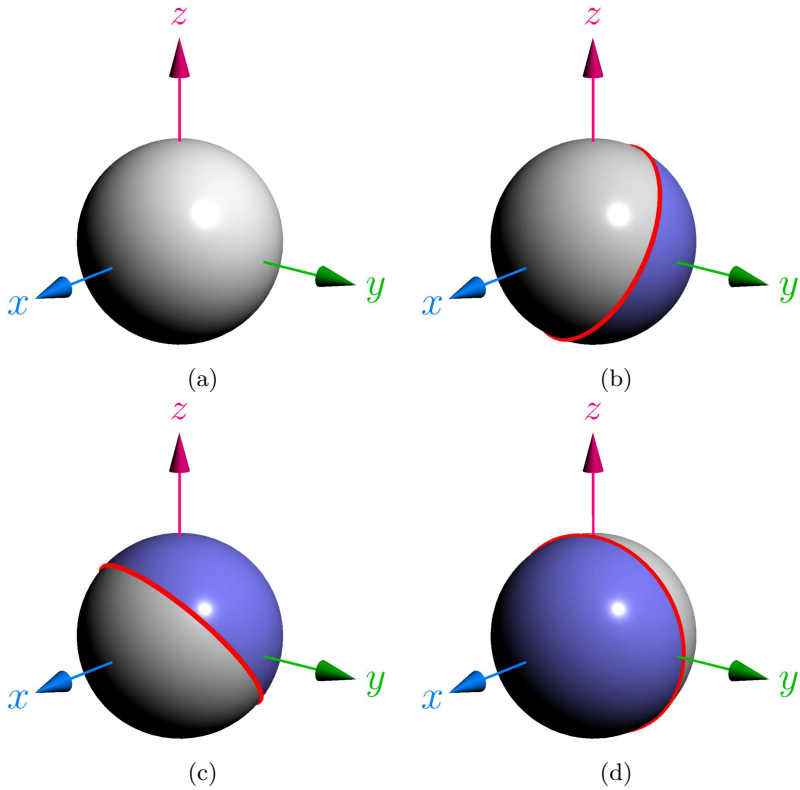


Figure 1: Putnam 2002 A2, Exercise 6

Summary — Apply the pigeonhole principle.

Walkthrough —

(a) Draw a great circle passing through at least two of the five points.

(b) At least one closed hemisphere contains at least two of the remaining three points.

(c) Conclude!

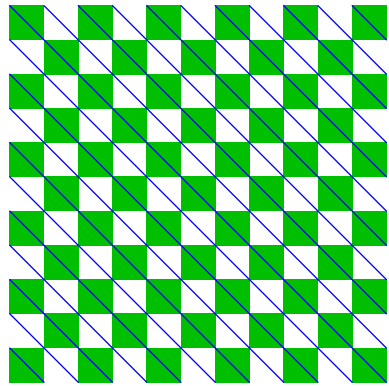
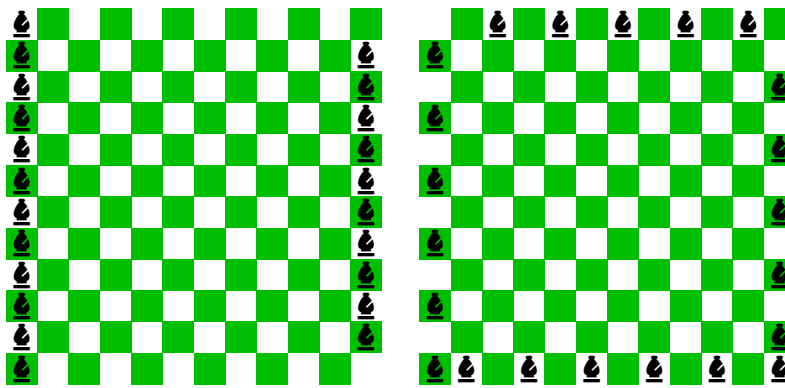


Figure 2: India BMath 2006

Solution 4. Draw a great circle passing through at least two of the five points. Then at least one closed hemisphere contains at least two of the remaining three points. This proves the result. See [**AN10**, Example 3.2]. ■

Exercise 7 (BMath 2006)**.** [India BMath 2006] Show that the maximum number of non-attacking bishops that can be put in an $n \times n$ chessboard is $2n - 2$.

> **Walkthrough** — Does considering a diagonal and the lines parallel to it help? See Fig. 2.



(a) (b)

Figure 3: India BMath 2006

Solution 5. The maximum number of non-attacking bishops that can be put in an $n \times n$ chessboard cannot exceed the number of diagonals (see Fig. 2) since

each square of the chessboard passes through at least one diagonal and each diagonal can contain at most one bishop. Moreover, the maximum number of non-attacking bishops is one less than the number of diagonals. Otherwise, each diagonal has to accommodate at least one bishop and then the bishops at the top-right corner and at the bottom-left corner would be attacking. If we put bishops at all the squares along the left column and the right column except the top-right and the bottom right squares (as in Fig. 3a), then these bishops would be non-attacking. Hence the maximum number of non-attacking bishops that can be put in an $n \times n$ chessboard is equal to $(2n-1)-1 = 2n-2$. ■

Exercise 8 (RMO 2011a P2,). Let $(a_1, a_2, a_3, \ldots, a_{2011})$ be a permutation of the numbers $1, 2, 3, \ldots, 2011$. Show that there exist two numbers j, k such that $1 \le j < k \le 2011$ and $|a_j - j| = |a_k - k|$.

Solution 6. Since $a_1, a_2, \ldots, a_{2011}$ is a permutation of $1, 2, \ldots, 2011$, the sum $\sum_{j=1}^{2011}(a_j - j)$ is equal to zero, which gives

$$\sum_{\substack{1\le j\le 2011\\ a_j - j > 0}} (a_j - j) = -\sum_{\substack{1\le j\le 2011\\ a_j - j < 0}} (a_j - j). \tag{1}$$

For any $1 \le j \le 2011$, note that $-2010 \le a_j - j \le 2010$, that is, $0 \le |a_j - j| \le 2010$ holds. If $|a_j - j|, |a_k - k|$ are equal for no j, k with $1 \le j < k \le 2011$, then $|a_1 - 1|, |a_2 - 2|, \ldots, |a_{2011} - 2011|$ are equal to $0, 1, 2, \ldots, 2010$ in some order, and hence

$$|a_1 - 1| + |a_2 - 2| + \cdots + |a_{2011} - 2011| = 1 + 2 + \cdots + 2010 = 1005 \cdot 2011,$$

which is odd and thus contradicts Eq. (1). So there exist two numbers j, k such that $1 \le j < k \le 2011$ and $|a_j - j| = |a_k - k|$. ■

Exercise 9 (RMO 2013d P6,). Suppose that the vertices of a regular polygon of 20 sides are coloured with three colors red, blue and green, such that there are exactly three red vertices. Prove that there are three vertices A, B, C of the polygon having the same colour such that triangle ABC is isosceles.

> **Walkthrough** — Decompose the set of vertices of the 20-gon using the vertices of the pentagons as in Fig. 4. Apply the pigeonhole principle. It would be useful to note that any three vertices of a regular pentagon form the vertices of an isosceles triangle.

Solution 7. Since there are exactly three red vertices and any of the remaining 17 vertices are blue or green, it follows that at least 9 of these 17 vertices are of the same color, say blue. Note that the set of vertices of a regular 20-gon can be written as the union of the four pairwise disjoint sets, each of them

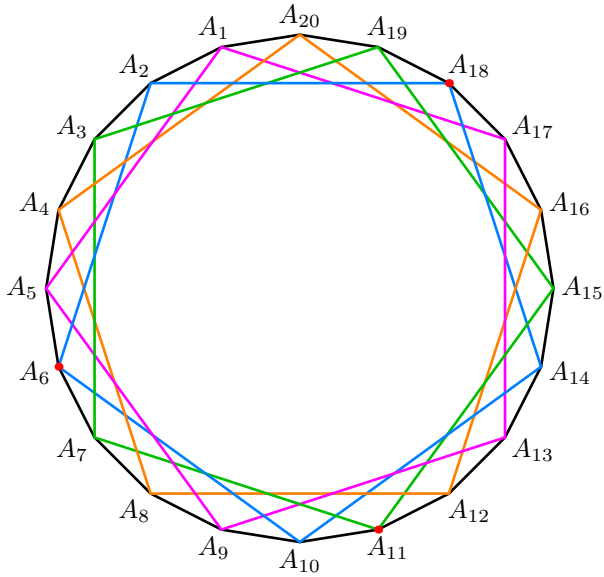


Figure 4: RMO 2013d P6, Exercise 9

consisting of the vertices of a regular pentagon (as in Fig. 4). Since there are nine blue vertices, by the pigeonhole principle, at least one of these four sets contains three blue points. Since any three points on a regular pentagon form the vertices of an isosceles triangle, the statement follows. ■

Example 10. [AE11, Problem 3.10] Show that the numbers 1 to 81 cannot be arranged in a 9×9 chessboard so that the product of the entries of the i-th row is equal to the product of the entries of the i-th column for any i, $1 \leq i \leq 9$.

> **Walkthrough** — Show that the diagonal contains the primes larger than $\frac{81}{2}$. How many primes are there between $\frac{81}{2}$ and 81?

Solution 8. Note that 41 occurs in certain row, say at the i-th row. So 41 divides the product of the entries of the i-th column. Since no number between 1 and 81 is a multiple of 41 except itself, 41 is common to i-th row and the i-th column, that is, it appears on the diagonal. Similarly, the diagonal contains all the primes less than 81 which are larger than 40, that is, it contains $41, 43, 47, 53, 59, 61, 67, 71, 73, 79$. However, 10 primes cannot be put along the diagonal. This proves the result. ■

Exercise 11 (RMO 2014a P4,). Is it possible to write the numbers $17, 18, 19, \ldots, 32$ in a 4×4 grid of unit squares with one number in each square

such that if the grid is divided into four 2×2 subgrids of unit squares, then the product of numbers in each of the subgrids divisible by 16?

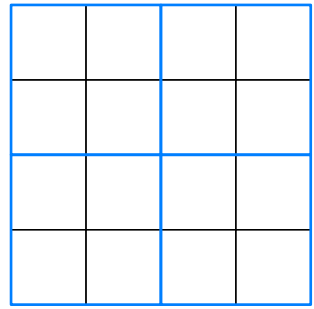


Figure 5: RMO 2014a P4, Exercise 11

Walkthrough —

(a) Note that the product of the entries in some subgrid is divisible by 32.

(b) Conclude that the product of all the 16 entries is divisible by $16 \times 16 \times 16 \times 32$.

(c) Is the product of the integers $17, 18, \ldots, 32$ divisible by $16 \times 16 \times 16 \times 32$?

Solution 9. The highest exponents of 2 dividing 32! and 16! are given by

$$v_2(32!) = 16 + 8 + 4 + 2 + 1, \quad v_2(16!) = 8 + 4 + 2 + 1.$$

So the highest power of 2 dividing the product of the integers $17, 18, 19, \ldots, 32$ is 2^{16}. Now if it were possible to write these numbers in a 4×4 grid in the above-mentioned manner, then the product of the numbers in each of the 2×2 subgrids with blue boundary (see Fig. 5) would be divisible by 2^4. Note that one such subgrid would contain 32, which implies that the product of $17, 18, \ldots, 32$ is divisible by $2^4 \cdot 2^4 \cdot 2^4 \cdot 32 = 2^{17}$, which is impossible. Hence it is not possible to write the integers $17, 18, \ldots, 32$ in a 4×4 grid satisfying the given conditions. ■

Exercise 12 (INMO 2016 P4,). Suppose 2016 points of the circumference of a circle are coloured red and the remaining points are coloured blue. Given any natural number $n \geq 3$, prove that there is a regular n-sided polygon all of whose vertices are blue.

Solution 10. Draw a regular n-gon circumscribed in the circle. If all its vertices are blue, then we are done. Otherwise, we rotate the regular n-gon by a sufficiently small angle so that all its vertices become red. This is possible since there are only finitely many blue points on the circle. ■

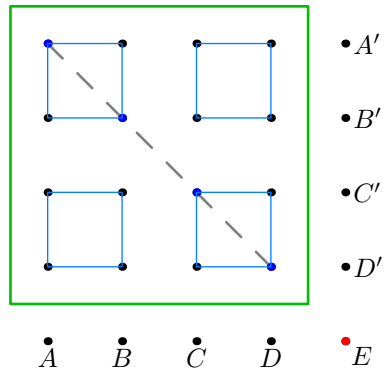


Figure 6: RMO 2018a P4, Exercise 13

Exercise 13 (RMO 2018a P4,). Let E denote the set of 25 points (m, n) in the xy-plane, where m, n are natural numbers, $1 \leq m \leq 5, 1 \leq n \leq 5$. Suppose the points of E are arbitrarily coloured using two colours, red and blue. Show that there always exist four points in the set E of the form $(a, b), (a + k, b), (a + k, b + k), (a, b + k)$ for some positive integer k such that at least three of these four points have the same colour. (That is, there always exist four points in the set E which form the vertices of a square and having at least three points of the same colour.)

Walkthrough —

(a) Assume that the conclusion is false, and by interchanging the colors (if necessary), assume that there are more red points than the blue ones.

(b) Show that one of the four corners is red, and assume without loss of generality that the point E (as in Fig. 6) is red.

(c) Considering the number of red points in each of the sets $\{A, A'\}$, $\{B, B'\}$, $\{C, C'\}$, $\{D, D'\}$, prove that these four sets together contain at most 4 red points, and conclude that the green square (as in Fig. 6) contains at least 8 red points.

(d) Consider the blue squares in the green square, to show that the green square contains exactly 8 red points.

(e) Conclude that there are exactly 13 red points, and each of the sets $\{A, A'\}$, $\{B, B'\}$, $\{C, C'\}$, $\{D, D'\}$ contains exactly one red point, and the green square contains exactly 8 red points.

(f) For each point on the dashed diagonal (as in Fig. 6), consider the square having this point and E as the endpoints of one of its diagonals to show that the points on the dashed diagonal are blue.

(g) Note that each point within the green square lying outside the dashed diagonal, is a vertex of a square having two of its remaining vertices on

the dashed diagonal. Conclude that each such point is red.

(h) Obtain a contradiction to conclude that the assumption is false.

Solution 11. On the contrary, let us assume that there is no axes-parallel square having at least three vertices of the same color.

Note that at least 13 among those 25 points are of the same color. Without loss of generality, assume that those are red. By our hypothesis, it follows that among the four vertices at the corners, there is at least one red vertex. Without loss of generality[1], let us assume that the bottom-right vertex is red, and denote this vertex by E (as in Fig. 6).

Note that each of the sets $\{A, A'\}, \{B, B'\}, \{C, C'\}, \{D, D'\}$ (as in Fig. 6) contains at most one red point, otherwise, we can form an axes-parallel square with at least three vertices of the same color. Consequently, the green square (as in Fig. 6) contains at least $13 - 5 = 8$ red points.

If the green square contains at least 9 red points, then at least one of the four blue squares (as in Fig. 6) contains at least three red points, which is not the case by our hypothesis. Hence, the green square contains exactly 8 red points. Consequently, there are precisely 13 red points among the 25 points, and each of the sets $\{A, A'\}, \{B, B'\}, \{C, C'\}, \{D, D'\}$ contains exactly one red point.

For any given point on the dashed diagonal (as in Fig. 6), consider the square having this point and E as the endpoints of one of its diagonals. Note that one of the other two vertices of this square is red. By our hypothesis, it follows that all the points on the gray diagonal (as in Fig. 6) are blue.

Note that each point within the green square lying outside the dashed diagonal, is a vertex of a square having two of its remaining vertices on the dashed diagonal. This shows that all the points within the green square, lying outside the gray diagonal, are red. Hence, there exists an axes-parallel square, all whose vertices are red. This contradicts our hypothesis, and hence, it follows that there exists an axes-parallel square having at least three vertices of the same color. ■

Exercise 14 (RMO 2023b P6,). Consider a set of 16 points arranged in a 4×4 square grid formation. Prove that if any 7 of these points are coloured blue, then there exists an isosceles right-angled triangle whose vertices are all blue.

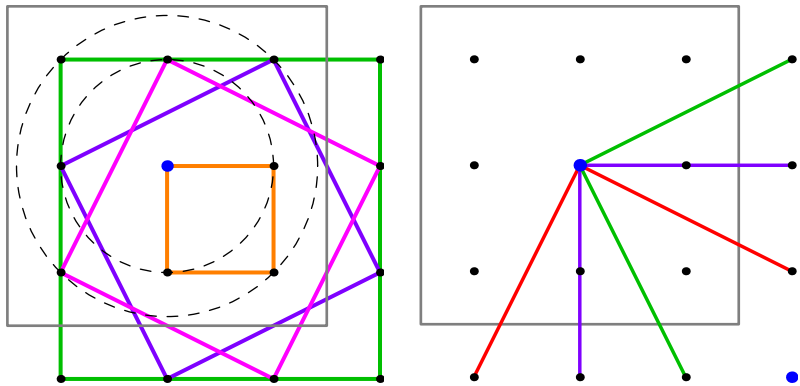
The following is from AoPS, is due to Rohan (Goyal?) as mentioned here by L567.

Walkthrough —

(a) Show that if the small square (as in Fig. 7a) does not contain a blue point, then we are done.

(b) It remains to consider the case when the small square (as in Fig. 7a)

[1] Convince yourself that there is **indeed no loss of generality** in assuming so.



(a) (b)

Figure 7: RMO 2023b P6, Exercise 14

contains at least one blue point.

(c) Rotating the configuration about the center of the small square (if necessary), assume that the top-left vertex of the small square (as in Fig. 7a) is blue.

(d) Prove that the gray square contains at most three blue points.

- Consider the case when at least two blue points lie on the bigger dashed circle. Show that the smaller dashed circle does not contain any blue point, in this case. Hence, the gray square contains at most three blue points.
- Similarly, if the smaller dashed circle contains at least two blue points, then the gray square (as in Fig. 7b) contains at most three blue points.

(e) It suffices to consider that each one of the red, purple, and green L-shapes, has at most one end-point which is blue (otherwise, we are done).

(f) Use the above to show that the bottom-right point (as in Fig. 7b) is blue.

(g) Consider the point at the bottom-right corner, and the center of the gray square, and a blue end-point of an L-shape, to show that these three points form the vertices of an isosceles triangle having the required properties.

Solution 12. Note that the 16 points are the vertices of the four squares as in Fig. 7a. If no vertex of the small square is blue, then by the pigeonhole principle, at least one of the remaining three squares has at least three blue vertices, and hence there exists an isosceles right-angled triangle with blue vertices.

Let us assume that the small square (as in Fig. 7a) has a blue vertex. Rotating the configuration about the center of the small square (if necessary), we may

and do assume that the top-left vertex of the small square is blue. Henceforth, on the contrary, let us assume that there are no isosceles right-angled triangles with blue vertices.

> **Claim** — The gray square (as in Fig. 7a) contains at most three blue points.

Proof of the Claim. Note that the points within the gray square lies on the two dashed circles (as in Fig. 7a). Therefore, to prove the Claim, it suffices to show that if one of the dashed circles (as in Fig. 7a) contains at least two blue points, then the other dashed circle does not contain any blue point.

If at least two blue points lie on the bigger dashed circle, then using the assumption, it follows that no more blue points lies on it, and hence these two blue points lie along a diameter. It follows that no blue point lies on the other dashed circle.

If at least two blue points lie on the smaller dashed circle, then using a similar argument, it follows that no blue point lies on the bigger dashed circle.

The Claim follows. □

Note that if both the end-points of one of the red, purple, and green L-shapes (as in Fig. 7b) are blue, then these points together with the center of gray square form the vertices of an isosceles right-angled triangle, contradicting the assumption. Hence, each one of these three L-shapes has at most one end-point which is blue. Since there are 7 blue points, using the Claim, it follows that the bottom-right point is blue. Note that the center of the gray square, the bottom-right point, and a blue end-point of the "purple" L-shape, form the vertices of a triangle having the required properties. ■

Example 15. Given five integers, show that the sum of some three of them is divisible by 3.

> **Walkthrough** —
>
> **(a)** If there are three integers among them leaving $0, 1, 2$ as remainders when divided by 3, then we are done.
>
> **(b)** Otherwise, there are three integers among them which leave the same remainder when divided by 3.

Solution 13. If each of the integers $0, 1, 2$ is congruent to at least one of the given integers, then we could take three integers among the five given integers, which are congruent to $0, 1, 2$ modulo 3, respectively, and note that their sum is divisible by 3.

Otherwise, there is a two-element subset A of $\{0, 1, 2\}$ such that each of those five integers is congruent to one of the elements of A modulo 3. Since

A contains two elements, by the pigeonhole principle, it follows that at least three of the given five integers are congruent to one of the elements in A. The sum of these three integers is divisible by 3. ■

Exercise 16 (RMO 2024a P6,). Let X be a set of 11 integers. Prove that one can find a nonempty subset $\{a_1, a_2, \ldots, a_k\}$ of X such that 3 divides k and 9 divides the sum $\sum_{i=1}^{k} 4^i a_i$.

> **Walkthrough** — Use repeatedly the fact that given five integers, the sum of some three of them is divisible by 3.

Solution 14. From the eleven elements of X, one can find three elements whose sum is divisible by 3. From the remaining 8 elements, one can find three elements whose sum is divisible by 3. From the remaining 5 elements, one can find three elements whose sum is divisible by 3. Hence, there are nine elements in X, denoted by $a_1, a_2, \ldots, a_9$ such that the sums

$$a_1 + a_2 + a_3, a_4 + a_5 + a_6, a_7 + a_8 + a_9$$

are divisible by 3. It follows that the sums

$$\alpha := \sum_{i=1}^{3} 4^i a_i, \beta := \sum_{i=4}^{6} 4^i a_i, \gamma := \sum_{i=7}^{9} 4^i a_i$$

are divisible by 3. Each of α, β, γ is congruent to one of $0, 3, 6$ modulo 9. To conclude, let us consider the following cases.

Suppose one of α, β, γ is divisible by 9. Noting that $4^3 \equiv 1 \pmod 9$, we can find a subset of X of size 3 with the desired property. Henceforth, let us assume that none of α, β, γ is divisible by 9.

If all of α, β, γ are congruent to 3 modulo 9, then we can find a subset of X of size 9 with the desired property.

If one of α, β, γ is congruent to 6 modulo 9, and the remaining two are congruent to 3 modulo 9, then we can find a subset of X of size 6 with the desired property.

If one of α, β, γ is congruent to 3 modulo 9, and the remaining two are congruent to 6 modulo 9, then we can find a subset of X of size 6 with the desired property.

If all of α, β, γ are congruent to 6 modulo 9, then we can find a subset of X of size 9 with the desired property.

The completes the proof. ■

Exercise 17. [Mat24, Problem 8, §6.4] Show that given 17 integers, the sum of some 9 of them is divisible by 9.

Walkthrough — Use repeatedly the fact that given five integers, the sum of some five of them is divisible by 3.

Exercise 18 (RMO 2019b P4,). Let $a_1, a_2, \ldots, a_6, a_7$ be seven positive integers. Let S be the set of all numbers of the form $a_i^2 + a_j^2$ where $1 \leq i < j \leq 7$. Prove that there exist two elements of S which have the same remainder on dividing by 36.

Solution 15. Note that the square of an integer is congruent to one of $0, 1, 4, 7$ modulo 9. It follows that the sum of two squares is congruent to one of $0, 1, 2, 4, 5, 7, 8$ modulo 9. Denote the set $\{0, 1, 2, 4, 5, 7, 8\}$ by R. Note that the sum of two squares is congruent to one of $0, 1, 2$ modulo 4. Let P denote the set of pairs of the form (i, j) with $1 \leq i < j \leq 7$.

If for some $s \in \{0, 1, 2\}$, there are at least eight pairs (i, j) in P such that the corresponding sums are congruent to s modulo 4, then by the pigeonhole principle, the result follows.

Suppose this does not hold. Noting that $\binom{7}{2} = 21 = 3 \cdot 7$, it follows that for each $s \in \{0, 1, 2\}$, there are precisely seven pairs in P such that the corresponding sums are congruent to s modulo 4.

In particular, there are seven pairs (i, j) in P such that the corresponding sums are divisible by 4, and hence, for any such pair, the integers a_i, a_j are even. If precisely k among the integers $a_1, a_2, \ldots, a_7$ are even, then there are $\binom{k}{2}$ pairs in P such that the corresponding sum is divisible by 4. We obtain $\binom{k}{2} = 7$, which holds for no integer $k \geq 0$. This completes the proof. ■

Exercise 19 (Danube Mathematical Competition, Junior 2019 P2,). Prove that for any real numbers $a_1, a_2, \ldots, a_n$ with $n \in \mathbb{N}$, there exists a real number x such that the numbers $x + a_1, x + a_2, \ldots, x + a_n$ are all irrational.

Walkthrough —

(a) Under what condition on x, it is false that all of the numbers $x + a_1, x + a_2, \ldots, x + a_n$ are irrational?

(b) What can be said about the set of real numbers x such that at least one of the numbers $x + a_1, x + a_2, \ldots, x + a_n$ is rational?

Solution 16. For any $1 \leq i \leq n$, note that the set of real numbers x such that $x + a_i$ is rational is countable, since it is the set of real numbers of the form $r - a_i$ for some rational number r. Since the union of countably many countable sets is countable, the set of real numbers x such that at least one of the numbers $x + a_1, x + a_2, \ldots, x + a_n$ is rational is countable. Since the set of real numbers is uncountable, there exists a real number x such that all of the numbers $x + a_1, x + a_2, \ldots, x + a_n$ are irrational. ■

Solution 17. Let y be an irrational number such that for any $1 \leq m \leq n+1$, at least one of the numbers $my + a_1, my + a_2, \ldots, my + a_n$ is rational. By the pigeonhole principle, there exists $1 \leq r \leq n$ such that at least two of the numbers $iy + a_r, jy + a_r$ are rational for some $1 \leq i < j \leq n+1$. Then y is rational, a contradiction. Hence, for any irrational number y, there exists $1 \leq m \leq n+1$ such that all of the numbers $my + a_1, my + a_2, \ldots, my + a_n$ are irrational. ■

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